

August 18, 2026

Editor-in-Chief
SIAM Undergraduate Research Online (SIURO)
Society for Industrial and Applied Mathematics

Dear Editor-in-Chief,

Please find enclosed our manuscript entitled "**Cycle-Aware Forman-Ricci Curvature for Signed and Bipartite Networks**," which we are submitting for exclusive consideration of publication as a research article in *SIAM Undergraduate Research Online (SIURO)*.

About the Paper and Its Importance: In this manuscript, we present a novel Cycle-Aware mathematical framework that extends standard Forman-Ricci Curvature to effectively analyze bipartite and signed networks without succumbing to metric collapse. By utilizing structurally bounded 4-cycle paths instead of absent 3-cycles, we restore the ability to detect and quantify dense mesoscopic clustering within these complex graphs. We also developed an optimized Masked Sparse-Sparse Matrix Multiplication (SpSpMM) algorithm that computes this metric deterministically with $\mathcal{O}(\sum d(v)d_{max})$ bounded time complexity. When validated on massive datasets like MovieLens-1M and Amazon Product Reviews, our zero-parameter geometric metric achieves predictive accuracy comparable to state-of-the-art Graph Neural Networks (like LightGCN) at a fraction of the computational cost.

Originality Statement: We confirm that this manuscript represents original research. It has not been previously published in any peer-reviewed venue, nor is it currently under consideration for publication by any other journal.

Suggested Reviewers: Given the interdisciplinary nature of this paper spanning pure combinatorics and applied network science, we respectfully suggest the following potential reviewers who have relevant expertise:

- **Dr. Jürgen Jost** (Max Planck Institute for Mathematics in the Sciences): Expertise in discrete Ricci curvature, Forman-Ricci extensions, and spectral graph theory.
- **Dr. Areejit Samal** (Institute of Mathematical Sciences, Chennai): Expertise in complex networks, Ollivier-Ricci curvature, and Forman-Ricci applications to empirical data.
- **Dr. Fan Chung** (UC San Diego): Expertise in pure combinatorics, spectral graph theory, and Cheeger inequalities on graphs.

Thank you very much for your time and for considering our work for publication in *SIURO*. We

look forward to your feedback.

Sincerely,

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